

SRAVYA CHANDOLU

Software Engineer | AI Engineer

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949-686-7114 | Fullerton, California, USA (Open for relocation)

SUMMARY

Software Engineer specializing in building LLM-powered systems, deploying AI Agents, AI workflows, and multilingual chatbots. Experience evaluating LLM performance through A/B testing and various model tradeoffs. Passionate about exploring and applying new LLM tools and frameworks to high-impact, real-world use cases, particularly in mission-driven domains

EDUCATION

Master of Science in Computer Science, GPA: 3.9/4.0

January 2023 - January 2025

California State University, Fullerton | *Graduate Teaching Associate for Python Programming*

SKILLS

- **Programming:** Python, TypeScript, React, Node.js
- **Back-end development:** MySQL, PostgreSQL, SQL ORM, Microservices, API Development (REST, FastAPI)
- **DevOps & Cloud:** Docker, Kubernetes (KIND), AWS, GCP, Azure AI Foundry, CI/CD pipelines, Git
- **Generative AI:** LLM (OpenAI, Claude, Gemini), Agents, RAG, Frameworks (LangChain, LangGraph, Crew AI), Fine-tuning (SFT), PyTorch

EXPERIENCE

Software Engineer | Hilal Software Corporation (Startup) | California, USA

January 2024 - Present

- **Led** a product **recommendation system** design with **LangChain**, prompt engineering (few-shot), and vector DB (FAISS), reducing latency by 62%
- Built a **multi-agent** research prototype using **LangGraph** to automate research, boosting efficiency by 60% with **human-in-the-loop** validation
- Deployed 20+ **FastAPI** endpoints for emotion, sentiment, language translation, and document parsing to support real-time backend **NLP** tasks
- Developed **RAG** pipelines and **AI Agents** using state-of-the-art **NLP/foundational models**
- Enhanced **PostgreSQL DB** performance by 30% using SQL query enhancements (schema design) and **database optimization** techniques (Indexing)

Software Engineer Intern | Head Starter AI | California, USA

June 2024 - November 2024

- Architected full-stack **web services** such as a ticketing platform (**500+** events) with **TypeScript**, React, Tailwind CSS, and Next.js
- **Collaborated** with cross-functional teams on code reviews (**agile sprints**) and conducted sessions to debug and define the **product roadmaps**
- Designed a **Machine Learning** model for predictive analytics of academic performance outcomes (96% accuracy) using the **Random Forest** algorithm
- Engineered a **conversational AI** chatbot using **AWS Bedrock**, Claude, and OpenAI, achieving 99.9% uptime on **EC2** for 1,000+ daily user interactions

PROJECTS

[Healthcare analytics platform](#) | Distributed Computing, Cloud | Python, GCP, Spark:

- Architected a scalable data pipeline on **GCP (clusters)** with Spark, leveraging BigQuery for **parallel processing** of 12+ GB patient datasets
- Achieved 67% **prediction accuracy** using Random Forest Classification (ML algorithm) for cardiovascular risk assessment

[Scalable E-Commerce Backend Infrastructure](#) | Python, Flask, ORM, Redis:

- Built a **Flask** backend with relational database (**MySQL**) for an e-commerce website, with 20% faster response times using **Redis caching**
- Improved system availability to handle **500+ requests/minute** through **load balancing** (Nginx) and **horizontal scaling** during high-traffic spikes

Voice-Enabled Database Query Interface | Python, LangChain, Whisper:

- Developed a voice-based **AI workflow** that lets users **query databases** using **natural language**, reducing query effort by 70% for **non-technical** users
- Used the **Crew AI** Agent tool and **Azure AI Foundry's** OpenAI for text processing. Leveraged **LangSmith** for AI system performance monitoring
- Integrated OpenAI Whisper for ASR (speech-to-text) and TTS models (gTTS) to **fine-tune** voice responses, and deliver an end-to-end voice interface
- Achieved over **90% accuracy** in natural language to SQL conversion and **deployed** the system with PostgreSQL for production database queries

[Ideavault - AI-driven Customer-Centric Web Application](#) | TypeScript, Next.js, Firebase:

- Designed a web platform to create, collaborate, and share documents, supporting **50+ concurrent** users using **TypeScript**, Next.js, and RESTful APIs
- Integrated **AI features** such as image generation using Cloudflare and OpenAI's tool Dall-E (LLM), improving customer efficiency by 30%
- Implemented **server-side rendering** and optimized performance using Lazy Loading and async. Reduced **page load time** by 35%
- Containerized microservices with **Docker** and deployed to Kubernetes